



HYDROTHERMAL VEIN

Water deep inside the Earth can become hot enough to dissolve minerals from rocks. If the resulting solution seeps up toward the surface, it cools and deposits the minerals in cracks and cavities. These form narrow veins and crystal-lined cavities known as geodes.

Quartz geode is colored by impurities in the minerals

GLITTERING FACES

The faces of crystals are perfectly flat, because of the way the atoms form the crystal bond. Sometimes the faces have steplike raised surfaces, but each step is perfectly aligned with the others on that face. This is why these galena crystals glitter with reflected light, like miniature mirrors.



Step in face

GEMSTONES

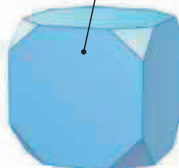
Transparent crystals, such as diamonds, bend light like tiny lenses, so that the crystal appears to sparkle. Gem-cutters enhance this effect by carefully reshaping the crystal. The resulting gemstone is smaller than the original crystal, but far more valuable.



SYMMETRICAL SHAPES

There are seven basic crystal types, which are usually grouped into six "families." Some are very regular, but others seem to have random shapes until you realize that their flat faces always lie at particular angles to each other. This feature is caused by their symmetry.

Cube has six faces



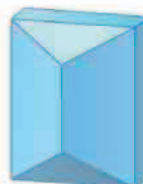
Cubic



Hexagonal or trigonal



Monoclinic



Orthorhombic



Tetragonal



Triclinic